car, your firecrackers, and you) contracted along the east–west direction, but not north–south or up–down. This was the contraction inferred by Fitzgerald, but now put on a firm foundation: The contraction is caused by the peculiar nature of space and time, and not by any physical forces that act on moving matter.

- Similarly, Einstein deduced that, as you speed eastward, you must see my space and everything at rest in it (my police station, my desk, and me) contracted along the east–west direction, but not north–south or up–down. That you see me contracted and I see you contracted may seem puzzling, but in fact it could not be otherwise: It leaves your state of motion and mine on an equal footing, in accord with the principle of relativity.
- Einstein also deduced that, as you speed past, I see your flow of time slowed, that is, dilated. The clock on your car's dashboard appears to tick more slowly than my clock on the police station wall. You speak more slowly, your hair grows more slowly, you age more slowly than I.

  Similarly, in accord with the principle of relativity, as you speed past me, you see my flow of time slowed. You see the clock on my station wall tick more slowly than the one on your dashboard. To you I seem to speak more slowly, my hair grows more slowly, and I age more slowly than you.

Box 1.1

Einstein's Proof of the Mixing of Space and Time

Einstein's principle of the absoluteness of the speed of light enforces the mixing of space and time; in other words, it enforces the relativity of simultaneity: Events that are simultaneous as seen by you (that lie in your space at a specific moment of your time), as your sports car speeds down Colorado Boulevard, are not simultaneous as seen by me, at rest in the police station. I shall prove this using descriptive words that go along with the spacetime diagrams shown below. This proof is essentially the same as the one devised by Einstein in 1905.